

CASE STUDY · AI AUTOMATION

AI Customer Support Agent Workflow

n8n + Ollama + Llama 3.2

An intelligent, intent-driven routing system that classifies inbound customer messages and dispatches auto-responses — zero human intervention for tier-1 support.

n8n

Ollama

Llama 3.2

Docker

Switch Node

Prompt Eng.

AI

OVERVIEW

Project Summary

This project implements an AI-powered customer support automation pipeline using **n8n** as the workflow orchestrator, **Ollama** as the local LLM server, and **Llama 3.2** as the language model. The system receives customer messages, classifies their intent using an LLM, and automatically routes each message to the correct response — eliminating manual triage for tier-1 support queries.

4 Intent Categories	100% Auto-Routed Responses	0 Human Intervention (Tier-1)	Local LLM Deployment
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SYSTEM ARCHITECTURE

End-to-End Pipeline

The workflow follows a linear, single-direction flow. Each node has a clearly defined responsibility, making the system easy to debug, extend, and migrate to other LLM providers.

1 Manual Trigger Initiates the workflow	2 Customer Message Input node — raw text	3 Basic LLM Chain Sends message to Ollama	4 Llama 3.2 Local model via Ollama	5 Intent Classification Returns category label	6 Switch Node Routes by intent	7 Auto Response Edit Fields node per route
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Each arrow represents a direct data handoff. The Switch Node routes based on the classified intent label returned by the LLM.

INTENT CLASSIFICATION

4 Route Categories & Auto-Responses

The LLM classifies every incoming message into exactly one of four intent categories. The Switch Node reads this label and dispatches to the corresponding Edit Fields node, which injects the pre-written response.

REFUND Customer requests reimbursement, return processing, or payment reversal. Auto-response: "Your refund request has been received and will be reviewed within 24 hours."	ORDER_STATUS Queries about delivery timeline, shipment tracking, or order progress. Auto-response: "Your order is currently being processed. Please check your tracking link for the latest status."
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COMPLAINT

Negative experience reports, product/service issues, or escalation requests.

Auto-response:

"We are sorry for the inconvenience. Your complaint has been registered and our support team will contact you shortly."

GENERAL

All other queries that do not match the three categories above.

Auto-response:

"Thank you for contacting us. Our support team is happy to assist you with your query."

WORKFLOW LOGIC

Step-by-Step Execution

Step 1

Receive customer message

The workflow is triggered manually or via webhook. The customer's raw text message is captured as the input payload.

Step 2

Send to Llama 3.2 via Ollama

The Basic LLM Chain node formats the message with a system prompt instructing the model to respond ONLY with one of the four intent labels.

Step 3

Classify into one category

Llama 3.2 processes the message and returns a single-word classification: REFUND, ORDER_STATUS, COMPLAINT, or GENERAL.

Step 4

Route via Switch Node

The Switch Node reads the LLM output and activates the matching branch. Each branch connects to a dedicated Edit Fields node.

Step 5

Return auto-response

The active Edit Fields node injects the pre-written response string. This value is returned as the workflow output.

TOOLS USED

Technology Stack

n8n	Workflow orchestration platform. Visual node-based automation with built-in LLM Chain support.
Docker	Containerizes both n8n and Ollama for consistent, reproducible local deployment.

Ollama	Runs LLMs locally. Acts as a lightweight inference server serving Llama 3.2.
Llama 3.2	Open-source LLM used for intent classification. Efficient, privacy-preserving, runs offline.
Basic LLM Chain	n8n node that formats the prompt and calls the Ollama API endpoint.
Switch Node	n8n node that branches workflow execution based on the classified intent label.
Edit Fields Nodes	One per intent route. Injects the pre-written auto-response into the output payload.

INTERVIEW TALKING POINTS

Key Topics to Present

- **AI Customer Support Automation**
Explain how the workflow eliminates manual tier-1 triage by using an LLM to classify and route messages automatically, reducing response time to near-zero.
 - **Intent Classification using LLMs**
Describe the prompt design: the system prompt instructs Llama 3.2 to return exactly one of four labels, making the output deterministic and easy to parse downstream.
 - **Prompt Engineering**
Highlight how constraining the model output to a fixed vocabulary (4 labels only) avoids parsing issues and makes the workflow robust without post-processing.
 - **Conditional Routing with Switch Nodes**
Explain that n8n's Switch Node evaluates the LLM output string and activates the correct branch — similar to a match/case statement in code.
 - **Local LLM Deployment using Ollama**
Emphasize privacy: no data leaves the machine. Ollama runs Llama 3.2 as a local REST API, making this suitable for sensitive enterprise use cases.
 - **Scalable Workflow Design in n8n**
Point out that adding a new intent category only requires: (1) updating the prompt, (2) adding a Switch branch, (3) adding one Edit Fields node — no code changes.
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MIGRATION PATH

This workflow was built and tested using **Llama 3.2 through Ollama**. It can be migrated to **Claude API (Anthropic)** by replacing the model node and configuring Anthropic API credentials. The routing logic, Switch Node, and Edit Fields nodes require **zero changes**.